

# Tutorial: Random Variables

## Solved Problems

## Theory: Random Variables and Probability Distributions

### 1. What is a Random Variable?

A random variable is a function that assigns a numerical value to each outcome in a sample space. It provides a mathematical way to describe uncertainty.

There are two main types of random variables:

### 2. Discrete Random Variables

A discrete random variable takes **countable values** (finite or countably infinite).

Examples:

- Number of heads when tossing coins
- Number of defective items in a batch

#### Probability Mass Function (PMF)

The PMF is defined as:

$$P(X = x)$$

and must satisfy:

1.  $P(X = x) \geq 0$
2.  $\sum_x P(X = x) = 1$

For discrete random variables:

$$P(X = a) \neq 0 \text{ in general}$$

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### 3. Continuous Random Variables

A continuous random variable takes **uncountably infinite values** over an interval.

Examples:

- Height of a person
- Time until a machine fails

#### Probability Density Function (PDF)

The PDF is denoted by:

$$f_X(x)$$

It satisfies:

1.  $f_X(x) \geq 0$
2.  $\int_{-\infty}^{\infty} f_X(x) dx = 1$

For continuous random variables:

$$P(X = a) = 0$$

Probabilities are computed over intervals:

$$P(a < X < b) = \int_a^b f_X(x) dx$$

### 4. Cumulative Distribution Function (CDF)

The CDF is defined for both discrete and continuous random variables as:

$$F_X(x) = P(X \leq x)$$

For discrete variables:

$$F_X(x) = \sum_{t \leq x} P(X = t)$$

For continuous variables:

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$

**Key Difference:**

|                        | Discrete     | Continuous   |
|------------------------|--------------|--------------|
| Function               | PMF          | PDF          |
| Probability at a point | Can be $> 0$ | Always 0     |
| Total probability      | Sum = 1      | Integral = 1 |
| Computation            | Summation    | Integration  |

## 5. Joint Distributions and Independence

For two random variables:

Discrete case:

$$P(X = x, Y = y)$$

Continuous case:

$$f_{X,Y}(x, y)$$

They are independent if:

$$f_{X,Y}(x, y) = f_X(x)f_Y(y)$$

(or the analogous PMF condition in the discrete case).

### Problem 1

Three fair coins are tossed. Let  $X$  denote the number of heads obtained.

- (a) List the sample space.
- (b) Obtain the probability mass function (PMF) of  $X$ .
- (c) Verify that it is a valid PMF.

### Solution

Sample space:

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}.$$

| Outcome       | Heads | $X$ |
|---------------|-------|-----|
| HHH           | 3     | 3   |
| HHT, HTH, THH | 2     | 2   |
| HTT, THT, TTH | 1     | 1   |
| TTT           | 0     | 0   |

PMF:

$$P(X = 0) = \frac{1}{8}, \quad P(X = 1) = \frac{3}{8}, \quad P(X = 2) = \frac{3}{8}, \quad P(X = 3) = \frac{1}{8}.$$

Verification:

$$\sum_x P(X = x) = 1.$$

## Problem 2

A random variable has PMF

$$P(X = x) = k(2x + 1), \quad x = 0, 1, 2.$$

- (a) Find  $k$ .
- (b) Compute  $P(X \geq 1)$ .

## Solution

Normalization:

$$k(1 + 3 + 5) = 1 \Rightarrow 9k = 1.$$

$$k = \frac{1}{9}.$$

$$P(X \geq 1) = \frac{3}{9} + \frac{5}{9} = \frac{8}{9}.$$

## Problem 3

The cumulative distribution function is

$$F_X(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 \leq x \leq 1 \\ 1, & x > 1. \end{cases}$$

- (a) Find the PDF.
- (b) Compute  $P(0.3 < X < 0.8)$ .

## Solution

PDF:

$$f_X(x) = \frac{d}{dx} F_X(x) = 2x, \quad 0 < x < 1.$$

$$P(0.3 < X < 0.8) = F_X(0.8) - F_X(0.3) = 0.8^2 - 0.3^2 = 0.55.$$

## Problem 4

A random variable  $X$  has probability density function

$$f_X(x) = ce^{-2x}, \quad x > 0.$$

Find:

- (a) the constant  $c$ ,
- (b) the cumulative distribution function (CDF),
- (c)  $P(X > 1)$ .

## Solution

**Important property of a PDF:** Any probability density function must satisfy

$$\int_{-\infty}^{\infty} f_X(x) dx = 1,$$

since total probability equals one.

**(a) Finding the constant  $c$**  Since the density is defined only for  $x > 0$ ,

$$\int_0^{\infty} ce^{-2x} dx = 1.$$

Taking the constant outside the integral,

$$c \int_0^{\infty} e^{-2x} dx = 1.$$

Using

$$\int e^{-ax} dx = \frac{-1}{a} e^{-ax},$$

we obtain

$$\int_0^{\infty} e^{-2x} dx = \left[ -\frac{1}{2} e^{-2x} \right]_0^{\infty}.$$

As  $x \rightarrow \infty$ ,  $e^{-2x} \rightarrow 0$ , hence

$$0 - \left( -\frac{1}{2} \right) = \frac{1}{2}.$$

Therefore,

$$c \cdot \frac{1}{2} = 1 \quad \Rightarrow \quad \boxed{c = 2}.$$

**(b) Finding the CDF** The cumulative distribution function is defined as

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(t) dt.$$

Because the PDF is zero for  $x < 0$ ,

$$F_X(x) = \begin{cases} 0, & x < 0, \\ \int_0^x 2e^{-2t} dt, & x \geq 0. \end{cases}$$

Integrating,

$$\int 2e^{-2t} dt = -e^{-2t}.$$

Applying limits,

$$F_X(x) = [-e^{-2t}]_0^x = -e^{-2x} + 1.$$

Hence,

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-2x}, & x \geq 0. \end{cases}$$

**Interpretation:** The CDF represents accumulated probability up to  $x$ . As  $x$  increases, the term  $e^{-2x}$  decreases, so the accumulated probability approaches 1.

**(c) Finding  $P(X > 1)$**  For a continuous random variable,

$$P(X > 1) = 1 - F_X(1).$$

Substituting the CDF,

$$F_X(1) = 1 - e^{-2}.$$

Therefore,

$$P(X > 1) = 1 - (1 - e^{-2}) = \boxed{e^{-2}}.$$

**Alternative verification:**

$$P(X > 1) = \int_1^{\infty} 2e^{-2x} dx = [-e^{-2x}]_1^{\infty} = e^{-2}.$$

## Problem 5

The joint probability density function is given by

$$f_{X,Y}(x, y) = k(x + y), \quad 0 < x < 1, \quad 0 < y < 1.$$

- (a) Find the value of  $k$ .
- (b) Obtain the marginal probability density function of  $X$ .
- (c) Determine whether  $X$  and  $Y$  are independent.

## Solution

**Conceptual Note:** A joint PDF must satisfy the normalization condition

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1,$$

meaning that the total probability over the entire support equals one.

Here, the support is the square region

$$0 < x < 1, \quad 0 < y < 1.$$

**(a) Finding the constant  $k$**  Apply the normalization condition:

$$\int_0^1 \int_0^1 k(x + y) dx dy = 1.$$

First integrate with respect to  $x$ :

$$\int_0^1 (x + y) dx = \left[ \frac{x^2}{2} + yx \right]_0^1 = \frac{1}{2} + y.$$

Now integrate with respect to  $y$ :

$$\begin{aligned} k \int_0^1 \left( \frac{1}{2} + y \right) dy &= k \left[ \frac{y}{2} + \frac{y^2}{2} \right]_0^1 \\ &= k \left( \frac{1}{2} + \frac{1}{2} \right) = k. \end{aligned}$$

Hence,

$$\boxed{k = 1}.$$

**(b) Marginal PDF of  $X$**  The marginal density of  $X$  is obtained by integrating the joint PDF over all possible values of  $Y$ :

$$f_X(x) = \int_0^1 f_{X,Y}(x, y) dy.$$

Substituting  $k = 1$ ,

$$\begin{aligned} f_X(x) &= \int_0^1 (x + y) dy \\ &= \left[ xy + \frac{y^2}{2} \right]_0^1 = x + \frac{1}{2}, \quad 0 < x < 1. \end{aligned}$$

Thus,

$$\boxed{f_X(x) = x + \frac{1}{2}, \quad 0 < x < 1.}$$

**Interpretation:** The marginal PDF represents the probability distribution of  $X$  alone, obtained by “averaging out” the effect of  $Y$ .

(c) **Independence of  $X$  and  $Y$**  Two random variables are independent if and only if

$$f_{X,Y}(x, y) = f_X(x)f_Y(y)$$

for all  $(x, y)$  in the support.

First compute the marginal of  $Y$ :

$$f_Y(y) = \int_0^1 (x + y) dx = \frac{1}{2} + y.$$

Now compute the product of marginals:

$$f_X(x)f_Y(y) = \left(x + \frac{1}{2}\right) \left(y + \frac{1}{2}\right).$$

Expanding,

$$xy + \frac{x}{2} + \frac{y}{2} + \frac{1}{4}.$$

However, the joint PDF is

$$f_{X,Y}(x, y) = x + y.$$

Since

$$x + y \neq xy + \frac{x}{2} + \frac{y}{2} + \frac{1}{4},$$

the equality condition fails.

Therefore,

$X$  and  $Y$  are not independent.

## Problem 6

Let  $X \sim U(0, 1)$  and define a new random variable

$$Y = 2X + 3.$$

(a) Find the PDF of  $Y$ .

## Solution

**Conceptual Note:** When a random variable undergoes a transformation, its probability density changes according to how the transformation stretches or compresses the sample space.

Since  $X \sim U(0, 1)$ ,

$$f_X(x) = 1, \quad 0 < x < 1.$$

(a) **PDF of  $Y$**  Given

$$Y = 2X + 3,$$

first express  $X$  in terms of  $Y$ :

$$X = \frac{Y - 3}{2}.$$

The transformation rule for continuous random variables is

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|.$$

Compute the derivative:

$$\frac{dx}{dy} = \frac{1}{2}.$$

Since  $f_X(x) = 1$ ,

$$f_Y(y) = 1 \cdot \frac{1}{2} = \frac{1}{2}.$$

Next determine the support:

$$0 < X < 1 \Rightarrow 3 < Y < 5.$$

Hence,

$$f_Y(y) = \frac{1}{2}, \quad 3 < y < 5.$$

## Problem 7

Explain why, for a continuous random variable,

$$P(X = a) = 0.$$

### Solution

**Key Idea:** Probabilities for continuous random variables are computed using areas under the PDF curve.

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx.$$

If we consider a single point,

$$P(X = a) = \int_a^a f_X(x) dx.$$

Because the limits are identical, the interval width is zero:

$$\int_a^a f_X(x) dx = 0.$$

Hence,

$$P(X = a) = 0.$$

**Interpretation:** A single point has zero area under a curve. Probability exists only over intervals, not individual values.

## Problem 8

Two players alternately throw two dice. Player A wins if the sum is 9, and Player B wins if the sum is 4. Player A starts first.

Find the probability that Player B wins.

### Solution

#### Step 1: Compute single-turn winning probabilities

Total outcomes when two dice are thrown = 36.

Sum equal to 9:

$$(3, 6), (4, 5), (5, 4), (6, 3)$$
$$p_A = \frac{4}{36} = \frac{1}{9}.$$

Sum equal to 4:

$$(1, 3), (2, 2), (3, 1)$$
$$p_B = \frac{3}{36} = \frac{1}{12}.$$

#### Step 2: Understand the game structure

B wins only if:

- A fails first, and
- B succeeds on the next turn,

or both fail repeatedly before B eventually succeeds.

Thus the process repeats in cycles:

$$(A \text{ fails}) \rightarrow (B \text{ fails})$$

until B finally wins.

This forms a geometric series.

#### Step 3: Construct probability series

Probability B wins on first opportunity:

$$(1 - p_A)p_B.$$

If both fail once, probability is

$$(1 - p_A)(1 - p_B),$$

and the game restarts.

Hence,

$$P(B) = (1 - p_A)p_B [1 + (1 - p_A)(1 - p_B) + ((1 - p_A)(1 - p_B))^2 + \dots].$$

**Step 4: Use geometric series formula** For  $|r| < 1$ ,

$$1 + r + r^2 + \dots = \frac{1}{1 - r}.$$

Let

$$r = (1 - p_A)(1 - p_B).$$

Then

$$P(B) = \frac{(1 - p_A)p_B}{1 - (1 - p_A)(1 - p_B)}.$$

**Step 5: Substitute values**

$$\begin{aligned} P(B) &= \frac{(8/9)(1/12)}{1 - (8/9)(11/12)} \\ &= \frac{8/108}{20/108} = \boxed{\frac{2}{5}}. \end{aligned}$$

## Problem 9 (Practical: Customer Support Tickets)

A small helpdesk receives a random number of support tickets per hour. Let  $X$  be the number of tickets received in one hour. The PMF is:

$$P(X = x) = \begin{cases} 0.10, & x = 0 \\ 0.25, & x = 1 \\ 0.35, & x = 2 \\ 0.20, & x = 3 \\ 0.10, & x = 4 \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Verify that this is a valid PMF.
- (b) Compute  $P(X \leq 2)$ .
- (c) Compute  $P(1 \leq X \leq 3)$ .
- (d) Find the CDF  $F_X(x)$  (write it as a piecewise function).

### Solution

**(a) Valid PMF check** A PMF must satisfy:

$$P(X = x) \geq 0 \quad \text{and} \quad \sum_x P(X = x) = 1.$$

All probabilities are nonnegative. Now sum:

$$0.10 + 0.25 + 0.35 + 0.20 + 0.10 = 1.00.$$

Hence it is a valid PMF.

(b) **Compute**  $P(X \leq 2)$

$$P(X \leq 2) = P(0) + P(1) + P(2) = 0.10 + 0.25 + 0.35 = 0.70.$$

(c) **Compute**  $P(1 \leq X \leq 3)$

$$P(1 \leq X \leq 3) = P(1) + P(2) + P(3) = 0.25 + 0.35 + 0.20 = 0.80.$$

(d) **CDF**  $F_X(x) = P(X \leq x)$  The CDF accumulates probability up to  $x$ . Since  $X$  only takes values 0, 1, 2, 3, 4:

$$F_X(x) = \begin{cases} 0, & x < 0 \\ 0.10, & 0 \leq x < 1 \\ 0.10 + 0.25 = 0.35, & 1 \leq x < 2 \\ 0.70, & 2 \leq x < 3 \\ 0.90, & 3 \leq x < 4 \\ 1, & x \geq 4. \end{cases}$$

## Problem 10 (Practical: Delivery Time Window)

A courier's delivery time (in hours) is a continuous random variable  $T$  with PDF:

$$f_T(t) = ct, \quad 0 < t < 2.$$

(a) Find the constant  $c$ .

(b) Find the CDF  $F_T(t)$ .

(c) Compute  $P(0.5 < T < 1.5)$ .

(d) Find the median delivery time  $m$  such that  $P(T \leq m) = 0.5$ .

### Solution

(a) **Find  $c$  using normalization** A PDF must satisfy:

$$\int_{-\infty}^{\infty} f_T(t) dt = 1.$$

Here support is  $0 < t < 2$ , so:

$$\int_0^2 ct dt = 1.$$

Compute the integral:

$$c \left[ \frac{t^2}{2} \right]_0^2 = c \cdot \frac{4}{2} = 2c = 1 \quad \Rightarrow \quad \boxed{c = \frac{1}{2}}.$$

**(b) Find the CDF** For  $t < 0$ ,  $F_T(t) = 0$ . For  $0 \leq t \leq 2$ :

$$F_T(t) = \int_0^t \frac{1}{2}u \, du = \frac{1}{2} \left[ \frac{u^2}{2} \right]_0^t = \frac{t^2}{4}.$$

For  $t \geq 2$ ,  $F_T(t) = 1$ . So:

$$F_T(t) = \begin{cases} 0, & t < 0 \\ \frac{t^2}{4}, & 0 \leq t \leq 2 \\ 1, & t \geq 2. \end{cases}$$

**(c) Compute**  $P(0.5 < T < 1.5)$  For continuous RVs:

$$P(a < T < b) = F_T(b) - F_T(a).$$

So:

$$P(0.5 < T < 1.5) = \frac{1.5^2}{4} - \frac{0.5^2}{4} = \frac{2.25 - 0.25}{4} = \frac{2}{4} = 0.5.$$

$$P(0.5 < T < 1.5) = 0.5.$$

**(d) Median**  $m$  Median satisfies:

$$F_T(m) = 0.5.$$

For  $0 \leq m \leq 2$ ,  $F_T(m) = m^2/4$ , hence:

$$\frac{m^2}{4} = 0.5 \Rightarrow m^2 = 2 \Rightarrow m = \sqrt{2}.$$

$$m = \sqrt{2} \approx 1.414 \text{ hours.}$$

## Problem 11 (Practical: App Login Attempts)

When users try to log in, the number of attempts  $X$  needed (including the successful one) has:

$$P(X = x) = \left(\frac{2}{3}\right)^{x-1} \left(\frac{1}{3}\right), \quad x = 1, 2, 3, \dots$$

(a) Compute  $P(X = 1)$  and interpret it.

(b) Compute  $P(X \leq 3)$ .

(c) Compute  $P(X > 4)$ .

(d) Find the CDF  $F_X(x)$  for real  $x$  (piecewise form).

## Solution

(a)  $P(X = 1)$

$$P(X = 1) = \left(\frac{2}{3}\right)^0 \left(\frac{1}{3}\right) = \frac{1}{3}.$$

Interpretation: probability the user succeeds on the first attempt is  $1/3$ .

(b)  $P(X \leq 3)$

$$P(X \leq 3) = P(1) + P(2) + P(3).$$

Compute each term:

$$P(1) = \frac{1}{3}, \quad P(2) = \left(\frac{2}{3}\right)^1 \frac{1}{3} = \frac{2}{9}, \quad P(3) = \left(\frac{2}{3}\right)^2 \frac{1}{3} = \frac{4}{27}.$$

Sum:

$$\frac{1}{3} + \frac{2}{9} + \frac{4}{27} = \frac{9}{27} + \frac{6}{27} + \frac{4}{27} = \frac{19}{27}.$$

$$\boxed{P(X \leq 3) = \frac{19}{27}.$$

(c)  $P(X > 4)$

$$P(X > 4) = 1 - P(X \leq 4).$$

First compute  $P(X \leq 4)$ :

$$P(4) = \left(\frac{2}{3}\right)^3 \frac{1}{3} = \frac{8}{81}.$$

So:

$$P(X \leq 4) = \frac{19}{27} + \frac{8}{81} = \frac{57}{81} + \frac{8}{81} = \frac{65}{81}.$$

Hence:

$$P(X > 4) = 1 - \frac{65}{81} = \frac{16}{81}.$$

$$\boxed{P(X > 4) = \frac{16}{81}.$$

(d) **CDF (piecewise)** Because  $X$  is discrete on  $\{1, 2, 3, \dots\}$ :

$$F_X(x) = P(X \leq x) = \begin{cases} 0, & x < 1, \\ P(X \leq \lfloor x \rfloor), & x \geq 1. \end{cases}$$

Also, a compact formula for integer  $n \geq 1$ :

$$P(X \leq n) = 1 - \left(\frac{2}{3}\right)^n$$

(since  $P(X > n) = (2/3)^n$ ). So:

$$\boxed{F_X(x) = \begin{cases} 0, & x < 1, \\ 1 - \left(\frac{2}{3}\right)^{\lfloor x \rfloor}, & x \geq 1. \end{cases}}$$

## Problem 12 (Practical: Two-Sensor System, Joint PDF)

Two sensors measure normalized signals  $(X, Y)$  inside the square  $0 < x < 1$ ,  $0 < y < 1$ . The joint PDF is:

$$f_{X,Y}(x, y) = 6xy, \quad 0 < x < 1, \quad 0 < y < 1.$$

- (a) Verify it is a valid joint PDF.
- (b) Find the marginals  $f_X(x)$  and  $f_Y(y)$ .
- (c) Compute  $P(X < 0.5, Y < 0.5)$ .
- (d) Are  $X$  and  $Y$  independent?

### Solution

(a) **Validity check** Need:

$$\int_0^1 \int_0^1 6xy \, dx \, dy = 1.$$

Integrate in  $x$  first:

$$\int_0^1 6xy \, dx = 6y \left[ \frac{x^2}{2} \right]_0^1 = 6y \cdot \frac{1}{2} = 3y.$$

Now integrate in  $y$ :

$$\int_0^1 3y \, dy = 3 \left[ \frac{y^2}{2} \right]_0^1 = \frac{3}{2} \neq 1.$$

So as written, it is **not normalized**. The correct constant must be found.

Let  $f_{X,Y}(x, y) = kxy$ . Normalize:

$$\int_0^1 \int_0^1 kxy \, dx \, dy = 1.$$

Compute:

$$\int_0^1 ky \left[ \frac{x^2}{2} \right]_0^1 dy = \int_0^1 ky \cdot \frac{1}{2} dy = \frac{k}{2} \left[ \frac{y^2}{2} \right]_0^1 = \frac{k}{4}.$$

Set equal to 1:

$$\frac{k}{4} = 1 \Rightarrow \boxed{k = 4}.$$

So the valid joint PDF is:

$$\boxed{f_{X,Y}(x, y) = 4xy, \quad 0 < x < 1, \quad 0 < y < 1.}$$

**(b) Marginals**

$$f_X(x) = \int_0^1 4xy \, dy = 4x \left[ \frac{y^2}{2} \right]_0^1 = 2x, \quad 0 < x < 1.$$

$$f_Y(y) = \int_0^1 4xy \, dx = 4y \left[ \frac{x^2}{2} \right]_0^1 = 2y, \quad 0 < y < 1.$$

So:

$$\boxed{f_X(x) = 2x, \quad 0 < x < 1; \quad f_Y(y) = 2y, \quad 0 < y < 1.}$$

**(c) Probability in a rectangle**

$$P(X < 0.5, Y < 0.5) = \int_0^{0.5} \int_0^{0.5} 4xy \, dy \, dx.$$

Inner integral:

$$\int_0^{0.5} 4xy \, dy = 4x \left[ \frac{y^2}{2} \right]_0^{0.5} = 4x \cdot \frac{0.25}{2} = 4x \cdot 0.125 = 0.5x.$$

Outer integral:

$$\int_0^{0.5} 0.5x \, dx = 0.5 \left[ \frac{x^2}{2} \right]_0^{0.5} = 0.5 \cdot \frac{0.25}{2} = 0.5 \cdot 0.125 = 0.0625.$$

$$\boxed{P(X < 0.5, Y < 0.5) = 0.0625 = \frac{1}{16}.}$$

**(d) Independence check** Check whether:

$$f_{X,Y}(x, y) = f_X(x)f_Y(y).$$

We have:

$$f_X(x)f_Y(y) = (2x)(2y) = 4xy,$$

which matches the joint density. Therefore:

$$\boxed{X \text{ and } Y \text{ are independent.}}$$

## Problem 13 (Practical: Mixed Delivery Modes)

A package arrives using one of two delivery modes:

- With probability 0.7, it uses *Express* and delivery time  $T$  has PDF  $f_E(t) = e^{-t}$  for  $t > 0$ .
- With probability 0.3, it uses *Standard* and delivery time  $T$  has PDF  $f_S(t) = \frac{1}{2}e^{-t/2}$  for  $t > 0$ .

- (a) Find the overall PDF  $f_T(t)$ .
- (b) Find the overall CDF  $F_T(t)$ .
- (c) Compute  $P(T < 2)$ .
- (d) Compute  $P(T > 3)$ .

## Solution

**(a) Overall PDF (mixture)** A mixture PDF is the weighted sum:

$$f_T(t) = 0.7f_E(t) + 0.3f_S(t).$$

So:

$$f_T(t) = 0.7e^{-t} + 0.3 \left( \frac{1}{2}e^{-t/2} \right) = 0.7e^{-t} + 0.15e^{-t/2}, \quad t > 0.$$

**(b) Overall CDF** Compute each CDF first.

Express:

$$F_E(t) = \int_0^t e^{-u} du = [-e^{-u}]_0^t = 1 - e^{-t}.$$

Standard:

$$F_S(t) = \int_0^t \frac{1}{2}e^{-u/2} du.$$

Let  $w = u/2$ ,  $du = 2dw$ :

$$F_S(t) = \int_0^{t/2} \frac{1}{2}e^{-w} \cdot 2dw = \int_0^{t/2} e^{-w} dw = 1 - e^{-t/2}.$$

Now mix:

$$F_T(t) = 0.7F_E(t) + 0.3F_S(t) = 0.7(1 - e^{-t}) + 0.3(1 - e^{-t/2}).$$

So:

$$F_T(t) = 1 - 0.7e^{-t} - 0.3e^{-t/2}, \quad t > 0,$$

and  $F_T(t) = 0$  for  $t \leq 0$ .

**(c)  $P(T < 2)$**

$$P(T < 2) = F_T(2) = 1 - 0.7e^{-2} - 0.3e^{-1}.$$

$$P(T < 2) = 1 - 0.7e^{-2} - 0.3e^{-1}.$$

**(d)  $P(T > 3)$**

$$P(T > 3) = 1 - F_T(3) = 0.7e^{-3} + 0.3e^{-3/2}.$$

$$P(T > 3) = 0.7e^{-3} + 0.3e^{-3/2}.$$